One-page reflection — ERP financial reporting take-home

What I chose and why. I implemented manual adjustment validation because an invalid journal must be stopped before it contaminates any downstream financial statement. I preferred deterministic `Decimal` arithmetic, approved-COA checks, and explicit `ACCEPTED` / `REVIEW` / `REJECTED` decisions to a convincing but untrustworthy LLM-generated statement. I added a read-only scan of the other company inputs to demonstrate how upstream data quality can block downstream processing.

What I found in the supplied data. Of 10 journals, seven passed implemented checks, two were rejected (`JE-002` unbalanced by $3,500; `JE-005` references unknown account `6315`), and one was quarantined for review (`JE-008`, a same-account $320,000 intercompany debit/credit). The current TB includes an unmapped `9999`, repeated `6310`, and multiple currencies; the FX file lacks a GBP period-end rate. The prior TB has legacy account `6905`. These issues are intentionally not silently corrected. Passing journal checks does not establish posting approval or statement correctness.

With three months. I would build versioned, entity-aware ledger storage and human approval workflows first; then ERP adapters, a policy-driven FX translation engine, reconciliation/lineage DAG, and a deterministic BS/P&L compiler. Cash flow and SOCIE would follow only after transaction and equity-rollforward metadata were available. An optional retrieval-grounded LLM would rank mapping candidates and draft reviewer explanations; a benchmark would measure accuracy, abstention and reviewer override rates.

Where it breaks at scale. The prototype is a single-process CSV/JSON utility, not a transactional service. It does not support thousands of concurrent accounts, entity-scoped mappings, cross-period approval histories, restatements, large IC elimination graphs, streaming ERP deltas, or row-to-cell lineage. Its trace hash detects content changes but is not a signed audit log. Date validation is minimal, and it does not validate the accounting substance or FX calculation behind a balanced journal.

AI tool use and judgment. I used ChatGPT to help decompose the requirements, draft the architecture and reflection, propose failure cases, and accelerate Python/test writing. I checked the supplied files and ran the validator rather than trusting generated claims. An important AI failure mode was the temptation to call the same-account intercompany journal a proven circular transaction: without entity/counterparty identifiers, that conclusion is not justified. I therefore classified it as review-required, not automatically invalid. No LLM API is used at runtime.

What is underestimated. The hardest problem is the meaning and governance of data, not adding debits and credits. Two equally balanced journals can have radically different accounting validity. COA changes, functional currency, legal-entity ownership, historical rates, period cutoffs and restatement policies all require documented decisions, provenance and accountable human approval.